

Proofs and Problem Solving

Week 6: The Integers

Notes based on Martin Liebeck's

A Concise Introduction to Pure Mathematics

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11. The Integers

This week we will study the integers, particularly divisibility, the highest common factor, and prime factorization.

These techniques are especially useful for solving *diophantine equations*, which are polynomial equations where we seek *integer* solutions. One of the most famous diophantine equation is

$$x^n + y^n = z^n.$$

It was famously claimed (without proof) by Fermat in about 1637 that there are solutions $x, y, z, n \in \mathbb{N}$ only if $n = 1$ or $n = 2$. It wasn't until 1994 when Andrew Wiles actually gave a proof.

This week's set of tools are also useful when we are trying to show that a diophantine equation *cannot* be solved. You have seen one such proof before: that $\sqrt{2}$ is irrational. Let's recall the proof briefly:

Proof. Suppose there was a rational number $\frac{m}{n}$ (where m and n are reduced in the sense that they have no common divisors apart from 1) so that $(\frac{m}{n})^2 = 2$, then $m^2 = 2n^2$. This means that m^2 is even. Then m must be even since, if instead $m = 2k + 1$ for some integer k , then $m^2 = (2k + 1)^2 = 4(k^2 + k) + 1$, which is odd, a contradiction. Hence, $m = 2k$ for some integer k , and so

$$2n^2 = m^2 = (2k)^2 = 4k^2$$

and so $n^2 = 2k^2$, and using a similar reasoning, we get that n must also be even, which is a contradiction since we assumed m and n had no common factors, but now we have shown they are both divisible by 2. ■

The important feature of the proof was to use the *coprimality* of m and n , that is, that they share no common divisors, as well as the fact that if m^2 is divisible by 2 (that is, if m^2 was even, then so was m). If we want to generalize this proof, we will need to develop these tools about divisibility and coprimality. Later, we will in fact show that $\sqrt{n}$ is rational exactly when $n = m^2$ for some integer m .

As motivation for the techniques we develop along the way, we will show how they can be used to solve various diophantine equations. As a special final application of these techniques, we'll also classify all *Pythagorean Triples*, that is, integers x, y, z so that

$$x^2 + y^2 = z^2.$$

If you like the material from this week, you might enjoy taking *Introduction to Number Theory* (4th year).

11.1 Remainders and divisibility

The following theorem is the starting point for our studies on divisibility.

Theorem 11.1 — The Remainder Theorem. Let $a \in \mathbb{N}$ and $b \in \mathbb{Z}$. There are unique integers $q \in \mathbb{Z}$ and $0 \leq r < a$ such that

$$b = qa + r.$$

Proof. We prove existence and uniqueness separately.

Existence: Let q be the largest integer for which $qa \leq b$. Let $r = b - qa \geq 0$. Then we must also have $r < a$, since if $r \geq a$, then $b - qa \geq a$, and so $b - (q + 1)a \geq 0$, which contradicts q being the largest integer for which $qa \leq b$. This shows the existence of a pair (q, r) satisfying the theorem.

Uniqueness: Assume that (q', r') is a pair of integers such that $b = q'a + r'$ and $0 \leq r' < a$. Then

$$0 = b - b = q'a + r' - qa - r = (q' - q)a + (r' - r),$$

and so

$$r' - r = (q - q')a.$$

Since $0 \leq r, r' < a$, we know $|r - r'| < a$. Thus,

$$a > |r - r'| = |(q - q')a| = |q - q'| \cdot a,$$

which implies that $1 > |q - q'|$. Since $|q - q'|$ is a nonnegative integer, it therefore follows that $q = q'$ and hence, $r = r'$. ■

Definition 11.1 For two integers a and b we say a *divides* b if there is an integer c so that $b = ac$. (Some authors will write $a|b$ for the sentence “ a divides b .”)

So for example, 2 divides 4, but 2 does not divide 3.

Lemma 11.2 If a divides b and b divides a , then $a = \pm b$.

Proof. Exercise. ■

11.2 The GCD and the Euclidean Algorithm

Definition 11.2 Let a and b be two nonzero integers. A *common divisor* or *common factor* of a and b is an integer d such that d divides both a and b . The *greatest common divisor* (AKA the *highest common factor*) of a and b is the *largest* common divisor of a and b . We denote this integer d by $\gcd(a, b)$ (you may also see the notation $\text{hcf}(a, b)$).

If $\gcd(a, b) = 1$, then we say that a and b are said to be *coprime* or *relatively prime*.

Thus a and b are coprime if and only if their only (positive) common divisor is 1.

Example 11.1 $\gcd(15, 45) = 15$, $\gcd(6, 15) = 3$, and $\gcd(17, 91) = 1$.

For a prime $p > 1$ and any integer n , we have:

$$\gcd(p, n) = \begin{cases} 1 & \text{if } p \text{ does not divide } n, \\ p & \text{if } p \text{ divides } n. \end{cases}$$

Example 11.2 We allow ourselves the flexibility to speak of the gcd of negative integers as well as positive ones, but we notice that

$$\gcd(a, b) = \gcd(a, -b) = \gcd(-a, b) = \gcd(-a, -b).$$

(So for instance $\gcd(-6, -27) = \gcd(6, 27) = 3$.) As a result, we will mostly focus on the gcd of natural numbers.

Note that any integer n with divisor d also has $-d$ as a divisor. Indeed, if $d|n$, there exists an integer m such that $n = md = (-m)(-d)$ and since $-m \in \mathbb{Z}$, $-d$ is also a divisor of n . Hence, $\gcd(a, b) > 0$.

Definition 11.3 Let a and b be integers. An *integer linear combination* of a and b is an integer of the form $ma + nb$ for some $m, n \in \mathbb{Z}$.

Notice that a linear combination of linear combinations is a linear combination. That is, if c and d are each linear combinations of a and b , then for some integers m, n, p , and q , we have $c = ma + nb$ and $d = pa + qb$. If we form a linear combination of c and d , we get

$$rc + sd = (rm + sp)a + (rn + sq)b,$$

which is a linear combination of a and b .

A little trick that we will use repeatedly is the observation that common divisors of two integers are also divisors of linear combinations of those integers:

Lemma 11.3 — Linear Combination Lemma. Let a, b , and d be integers. If d divides both a and b , then d divides every integer linear combination of a and b . In other words, for every pair of integers $m, n \in \mathbb{Z}$, the integer d also divides $ma + nb$.

Proof. Since d divides a and d divides b , there are integers s and t so that $a = sd$ and $b = td$. Hence

$$ma + nb = msd + ntd = (ms + nt)d$$

and so d divides $ma + nb$. ■

Example 11.3 Let $a, b \in \mathbb{Z}$. We claim that if there is a linear combination of a and b equal to 1, then a and b are coprime. Indeed, if $m, n \in \mathbb{Z}$ have the property that $ma + nb = 1$, then any common divisor d of a and b must also divide 1, by the Linear Combination Lemma. But then $d = 1$, so $\gcd(a, b) = 1$.

In particular, this means that n and $n + 1$ are coprime for every $n \in \mathbb{Z}$ since $(n + 1) - n =$

1 and $(n+1) - n$ is an integer linear combination of n and $n+1$.

11.2.1 Euclidean Algorithm

The *Euclidean Algorithm* is a step-by-step method for calculating the gcd for any two numbers, by exploiting the Remainder Theorem and the Linear Combination Lemma.

Let $a, b \in \mathbb{N}$ where $a < b$ and suppose we want to find $\gcd(a, b)$. We are going to construct a decreasing sequence of nonnegative integers, starting with b and a :

$$r_0 = b > r_1 = a > r_2 > \cdots > r_n > r_{n+1} = 0,$$

such that for each $k \in \mathbb{N}$, if $k \leq n$, then:

1. r_{k-1} is an integer linear combination of r_k and r_{k+1} , and
2. r_{k+1} is an integer linear combination of r_k and r_{k-1} .

If we can construct a sequence with these properties, then $\gcd(a, b)$ will be the natural number r_n .

To begin the iteration, we set $r_0 = b$ and $r_1 = a$. To construct the rest of the sequence, assume that for some $k \in \mathbb{N}$ we have constructed nonnegative integers $r_0 = b > r_1 = a > r_2 > \cdots > r_k$ with the two properties above.

By the Remainder Theorem, there exist integers q_{k+1} and r_{k+1} such that $0 \leq r_{k+1} < r_k$ and

$$r_{k-1} = q_{k+1}r_k + r_{k+1}.$$

Thus r_{k-1} is a linear combination of r_k and r_{k+1} . Also, as $r_{k+1} = r_{k-1} - q_{k+1}r_k$, r_{k+1} is a linear combination of r_k and r_{k-1} . Hence we have found a nonnegative integer r_{k+1} that satisfies the conditions above.

The sequence is nonnegative and strictly decreasing so, by the Monotone Convergence Theorem, this process must terminate with $r_{n+1} = 0$ for some n . This completes the construction of the sequence

$$r_0 = b > r_1 = a > r_2 > \cdots > r_n > r_{n+1} = 0,$$

with the desired properties.

Let us now see why this algorithm gives us the gcd of a and b . The key is the Linear Combination Lemma.

Claim:

Given a sequence as above, $r_n = \gcd(a, b)$.

Proof. We first claim that $\gcd(a, b) \leq r_n$. Note that $r_0 = b$ and $r_1 = a$ are certainly linear combinations of a and b . If r_{k-1} and r_k are each integer linear combinations of a and b , so is r_{k+1} . By induction, we conclude that r_k is an integer linear combination of a and b for all integers k where $1 \leq k \leq n$. In particular, r_n is a linear combination of a and b . Hence, it follows from the Linear Combination Lemma that $\gcd(a, b)$ divides r_n . In particular, since both $\gcd(a, b)$ and r_n are nonnegative, this means that $\gcd(a, b) \leq r_n$.

On the other hand, we claim that $r_n \leq \gcd(a, b)$. It is enough to show that r_n is a common divisor of a and b since, by definition of the $\gcd(a, b)$, all common divisors must be at most $\gcd(a, b)$. We will show this by proving, by induction, that r_n divides r_k and r_{k-1} for all natural numbers $k \leq n$. Indeed, since $r_{n+1} = 0$ and r_{n-1} is a linear combination of r_n and r_{n+1} , it follows that $r_{n-1} = q_{n+1}r_n$ for some $q_{n+1} \in \mathbb{Z}$, so r_n divides r_{n-1} . Also, clearly, r_n divides itself. For the induction step, we assume that r_n divides r_{k+1} and r_k .

Since r_{k-1} is a linear combination of r_k and r_{k+1} , it follows from the Linear Combination Lemma that r_n divides r_{k-1} as well. So r_n divides both r_k and r_{k-1} . By induction, we conclude that r_n divides $r_1 = a$ and $r_0 = b$. Thus, r_n is a common divisor of a and b and must therefore satisfy $r_n \leq \gcd(a, b)$.

Hence, $\gcd(a, b) = r_n$. ■

Let's watch this algorithm in action.

Example 11.4 Let $a = r_1 = 6$ and $b = r_0 = 15$.

- We divide 15 by 6 to get $15 = 2(6) + 3$ with a remainder of $r_2 = 3$.
- We divide 6 by 3 to get $6 = 2(3) + 0$ with a remainder of $r_3 = 0$.

Thus $r_2 = 3$ is the gcd of 6 and 15.

Example 11.5 Let $a = r_1 = 48$ and $b = r_0 = 66$.

- We divide 66 by 48 to get $66 = 1(48) + 18$ with a remainder of $r_2 = 18$.
- We divide 48 by 18 to get $48 = 2(18) + 12$ with a remainder of $r_3 = 12$.
- We divide 18 by 12 to get $18 = 1(12) + 6$ with a remainder of $r_4 = 6$.
- We divide 12 by 6 to get $12 = 2(6) + 0$ with a remainder of $r_5 = 0$.

Thus $r_4 = 6$ is the gcd of 48 and 66.

Example 11.6 Let $a = r_1 = 120$ and $b = r_0 = 214$.

- We divide 214 by 120 to get $214 = 1(120) + 94$ with a remainder of $r_2 = 94$.
- We divide 120 by 94 to get $120 = 1(94) + 26$ with a remainder of $r_3 = 26$.
- We divide 94 by 26 to get $94 = 3(26) + 16$ with a remainder of $r_4 = 16$.
- We divide 26 by 16 to get $26 = 1(16) + 10$ with a remainder of $r_5 = 10$.
- We divide 16 by 10 to get $16 = 1(10) + 6$ with a remainder of $r_6 = 6$.
- We divide 10 by 6 to get $10 = 1(6) + 4$ with a remainder of $r_7 = 4$.
- We divide 6 by 4 to get $6 = 1(4) + 2$ with a remainder of $r_8 = 2$.
- We divide 4 by 2 to get $4 = 2(2) + 0$ with a remainder of $r_9 = 0$.

Thus $r_8 = 2$ is the gcd of 120 and 214.

Example 11.7 Let $a = r_1 = 14$ and $b = r_0 = 348$.

- We divide 348 by 14 to get $348 = 24(14) + 12$ with a remainder of $r_2 = 12$.
- We divide 14 by 12 to get $14 = 1(12) + 2$ with a remainder of $r_3 = 2$.
- We divide 12 by 2 to get $12 = 6(2) + 0$ with a remainder of $r_4 = 0$.

Thus $r_3 = 2$ is the gcd of 14 and 348.

Example 11.8 Let $a = r_1 = 2059$ and $b = r_0 = 6744$.

- We divide 6744 by 2059 to get $6744 = 3(2059) + 567$ with a remainder of $r_2 = 567$.
- We divide 2059 by 567 to get $2059 = 3(567) + 358$ with a remainder of $r_3 = 358$.
- We divide 567 by 358 to get $567 = 1(358) + 209$ with a remainder of $r_4 = 209$.
- We divide 358 by 209 to get $358 = 1(209) + 149$ with a remainder of $r_5 = 149$.
- We divide 209 by 149 to get $209 = 1(149) + 60$ with a remainder of $r_6 = 60$.
- We divide 149 by 60 to get $149 = 2(60) + 29$ with a remainder of $r_7 = 29$.
- We divide 60 by 29 to get $60 = 2(29) + 2$ with a remainder of $r_8 = 2$.
- We divide 29 by 2 to get $29 = 14(2) + 1$ with a remainder of $r_9 = 1$.

• We divide 2 by 1 to get $2 = 1(2) + 0$ with a remainder of $r_{10} = 0$.
Thus $r_9 = 1$ is the gcd of 2059 and 6744. In particular, 2059 and 6744 are coprime.

One important corollary of the Euclidean Algorithm is the following.

Theorem 11.4 — Bezout's Identity. Let a and b be integers. Then $\gcd(a, b)$ is an integer linear combination of a and b . That is, there are integers s and t such that $\gcd(a, b) = sa + tb$.

Let's revisit some of the examples above to find the s and t .

Example 11.9 Let $a = r_1 = 6$ and $b = r_0 = 15$.

- We divide 15 by 6 to get $15 = 2(6) + 3$ with a remainder of $r_2 = 3$.
- We divide 6 by 3 to get $6 = 2(3) + 0$ with a remainder of $r_3 = 0$.

Thus $\gcd(6, 15) = 3 = 15 - 2 \cdot 6$.

Example 11.10 Let $a = r_1 = 48$ and $b = r_0 = 66$.

- We divide 66 by 48 to get $66 = 1(48) + 18$ with a remainder of $r_2 = 18$.
- We divide 48 by 18 to get $48 = 2(18) + 12$ with a remainder of $r_3 = 12$.
- We divide 18 by 12 to get $18 = 1(12) + 6$ with a remainder of $r_4 = 6$.
- We divide 12 by 6 to get $12 = 2(6) + 0$ with a remainder of $r_5 = 0$.

Thus

$$\begin{aligned}\gcd(48, 66) &= 6 = 18 - 12 \\ &= 18 - (48 - 2 \cdot 18) \\ &= 3 \cdot 18 - 48 \\ &= 3 \cdot (66 - 48) - 48 \\ &= 3 \cdot 66 - 4 \cdot 48.\end{aligned}$$

Example 11.11 Let $a = r_1 = 2059$ and $b = r_0 = 6744$.

- We divide 6744 by 2059 to get $6744 = 3(2059) + 567$ with a remainder of $r_2 = 567$.
- We divide 2059 by 567 to get $2059 = 3(567) + 358$ with a remainder of $r_3 = 358$.
- We divide 567 by 358 to get $567 = 1(358) + 209$ with a remainder of $r_4 = 209$.
- We divide 358 by 209 to get $358 = 1(209) + 149$ with a remainder of $r_5 = 149$.
- We divide 209 by 149 to get $209 = 1(149) + 60$ with a remainder of $r_6 = 60$.
- We divide 149 by 60 to get $149 = 2(60) + 29$ with a remainder of $r_7 = 29$.
- We divide 60 by 29 to get $60 = 2(29) + 2$ with a remainder of $r_8 = 2$.
- We divide 29 by 2 to get $29 = 14(2) + 1$ with a remainder of $r_9 = 1$.
- We divide 2 by 1 to get $2 = 1(2) + 0$ with a remainder of $r_{10} = 0$.

Thus

$$\begin{aligned}
 \gcd(6744, 2059) &= 1 = 29 - 14 \cdot 2 \\
 &= 29 \cdot 29 - 14 \cdot 60 \\
 &= 29 \cdot 149 - 72 \cdot 60 \\
 &= 101 \cdot 149 - 72 \cdot 209 \\
 &= 101 \cdot 358 - 173 \cdot 209 \\
 &= 274 \cdot 358 - 173 \cdot 567 \\
 &= 274 \cdot 2059 - 995 \cdot 567 \\
 &= 3259 \cdot 2059 - 995 \cdot 6744
 \end{aligned}$$

Bezout's Identity is incredibly useful, even in examples that seem to have nothing to do with the gcd:

■ **Exercise 11.1** Let a and b be coprime integers. Show that

(i) If a divides c and b divides c then ab divides c .

(ii) If a divides bc then a divides c .

Show that both parts of this result can fail if a and b are not coprime.

■ **Exercise 11.2** Show that if a, b are positive integers, and $d = \gcd(a, b)$ then there exist positive integers s, t such that $d = sa - tb$.

11.3 Corollaries of Bezout's Identity and the Linear Combination Lemma

Below we prove some useful corollaries using Bezout's Identity (Theorem 11.4) and the Linear Combination Lemma.

Corollary 11.5 Let $a, b, c \in \mathbb{Z}$. Suppose $c \neq 0$, c divides ab and $\gcd(a, c) = 1$. Then c divides b .

Proof. By Bezout's Identity, there are integers s and t such that $1 = \gcd(c, a) = sa + tc$. This gives $b = sab + tcb$. As c divides ab , then c divides b by the Linear Combination Lemma. ■

Corollary 11.6 Let $a, b \in \mathbb{Z}$ and let $d \in \mathbb{Z}$ be such that d divides a and d divides b . Then d divides $\gcd(a, b)$.

Proof. By Bezout's Identity, there are integers s and t such that $\gcd(a, b) = sa + tb$. Then as d divides a and d divides b , we have d divides $\gcd(a, b)$ by the Linear Combination Lemma. ■

Corollary 11.7 Let d be a common divisor of a and b , which is divisible by all divisors of a and b . Then $d = \pm \gcd(a, b)$.

Proof. By the previous corollary, d divides $\gcd(a, b)$, and since d is divisible by all divisors, it is divisible by $\gcd(a, b)$, so now we apply Lemma 11.2. ■

Here are some versions of the above corollaries when some of the numbers involved are primes:

Corollary 11.8 If $a, b \in \mathbb{Z}$, p is prime, and p divides ab , then either p divides a or p divides b (or both).

This just follows from Corollary 11.5 by letting $c = p$.

Corollary 11.9 If $n = p_1^{m_1} \cdots p_k^{m_k}$, where each of $p_1, \dots, p_k$ is *prime*, and if p is a prime number that divides n , then $p = p_i$ for some $i = 1, \dots, k$.

Proof. We prove this by induction. If $k = 1$, then p divides $p_1^{m_1}$, so by the previous corollary, $p = p_1$. Assume that the theorem is true for $k = r$; let us prove it for $k = r + 1$. Assume that p divides $p_1^{m_1} \cdots p_{r+1}^{m_{r+1}}$. Again by the previous corollary, either p divides $p_{r+1}^{m_{r+1}}$ or p divides $p_1^{m_1} \cdots p_r^{m_r}$. Thus, either $p = p_{r+1}$, or, by the induction hypothesis, $p = p_i$ for some $i = 1, 2, \dots, r$. This completes the proof. ■

Let's use these results to solve a simple diophantine equation.

Example 11.12 Find all integer solutions to $2x = 5y$.

Suppose (x, y) are integers solving $2x = 5y$. Then Corollary 11.8 implies 2 divides y and 5 divides x , so we let $y = 2z$ and $x = 5w$ for some integers z and w . Inserting these into the original equation, we get a new equation

$$10w = 2(5w) = 5(2z) = 10z.$$

Thus, $w = z$. Hence, any solution (x, y) must be of the form $(x, y) = (5w, 2w)$ for some integer w . We can also see that each pair of integers of the form $(5w, 2w)$ is a solution. Thus, the solutions to $2x = 5y$ are exactly all pairs of integers $\{(5w, 2w) : w \in \mathbb{Z}\}$.

11.4 The Fundamental Theorem of Arithmetic (FTA)

The Fundamental Theorem of Arithmetic says that all integers have a *unique* factorization as a product of (positive) powers of **distinct** prime numbers. We call this unique factorization of an integer x the **prime decomposition** of x .

Theorem 11.10 — The Fundamental Theorem of Arithmetic (FTA). Let $n \geq 2$ be an integer.

- (Existence) Then n is equal to a product $n = p_1^{r_1} \cdots p_k^{r_k}$ of powers of prime numbers, where $p_1 < \cdots < p_k$ and $r_i > 0$ for all i .
- (Uniqueness) The factorization is unique. That is, if we also have

$$p_1^{r_1} \cdots p_k^{r_k} = n = q_1^{s_1} \cdots q_\ell^{s_\ell}$$

where $q_1 < \cdots < q_\ell$ are primes and $s_i > 0$, then $k = \ell$, $p_i = q_i$, and $r_i = s_i$ for all i .

We will split the proof into three lemmas:

Lemma 11.11 — Existence of prime decomposition, Part I. Every integer $n \geq 2$ can be written as a product of primes $n = p_1 \cdots p_k$.

Remark: If n is prime, this statement still makes sense: we just interpret n as being the product of just one number, n itself.

Proof. We prove this by strong induction on n . The case $n = 2$ immediately holds (taking into account the previous remark). For the induction step, suppose the theorem holds for all integers $n < N$. If N is prime, there is nothing to prove. Otherwise, if N is not prime, then $N = a \cdot b$ for some positive integers a and b , both greater than 1 and less than N . Both a and b can be decomposed by the strong induction hypothesis, thus so can $N = ab$. ■

Lemma 11.12 — Existence of prime decomposition, Part II. Every integer $n \geq 2$ can be written as a product of powers of primes $n = p_1^{r_1} \cdots p_k^{r_k}$ where $p_1 < \cdots < p_k$ and $r_i \geq 0$.

Proof. By the previous lemma, $n = q_1 \cdots q_\ell$ for some primes $q_1 \cdots q_\ell$ that are not necessarily distinct. If $p_1 < p_2 < \cdots < p_k$ are the *distinct* primes that appear in the list $q_1, \dots, q_\ell$, let r_i denote the number of times that p_i appears in the list. Then

$$n = q_1 \cdots q_\ell = p_1^{r_1} \cdots p_k^{r_k}$$

which proves the lemma. ■

Lemma 11.13 — Uniqueness of prime decomposition. Every integer n can be written as a *unique* product of powers of primes $n = p_1^{r_1} \cdots p_k^{r_k}$.

Proof. Suppose $p_1^{r_1} \cdots p_k^{r_k} = n = q_1^{s_1} \cdots q_\ell^{s_\ell}$ are two decompositions. By cancelling any common factors, we can assume that no p_i equals any q_j . If there are any p_i 's and q_j 's remaining, Corollary 11.9 implies each p_i equals some q_j , which is a contradiction, thus there can be no terms remaining, so the two factorizations must have been equal. ■

Let's use the FTA to solve another simple-looking diophantine equation.

Example 11.13 Find all integer solutions to $x^2 - y^2 = 91$.

Reductions: Let's make a few observations first to narrow down what to solve for.

- Notice that if (x, y) is a solution, then so is $(\pm x, \pm y)$, and so we can assume that $x, y \geq 0$, since then the other solutions will be of the form $(\pm x, \pm y)$.
- We can't have $x = y$ since then the equation is not satisfied, so assume $x \neq y$. We can also assume $x > y$, since $x < y$ would imply $x^2 < y^2$, so $x^2 - y^2 < 0 < 91$.
- Finally, we can't have either x or y equal to zero, since 91 is not a perfect square.

Thus, after these reductions, we can assume $x > y > 0$.

Note that 91 has prime factorization $91 = 7 \cdot 13$, and so

$$91 = 7 \cdot 13 = x^2 - y^2 = (x - y)(x + y).$$

Since $x + y > x - y$, there are two cases to consider:

- If $x + y = 91$ and $x - y = 1$, solving these linear equations simultaneously gives $x = 46$ and $y = 45$.
- If $x + y = 13$ and $x - y = 7$, solving these two equations gives $x = 10$ and $y = 3$.

Thus, all the *positive* solutions (x, y) are $(46, 45)$ and $(10, 3)$. Thus, recalling our reductions, *all* solutions are just $(\pm 46, \pm 45)$, and $(\pm 10, \pm 3)$ (where we range over all possible combinations of $\pm$ for a total of 8 solutions).



Reductions: As we did in the example above, look for ways of simplifying your problem from the start by reducing the number of cases you have to investigate.

11.5 Finding divisors via FTA

If we know the prime decompositions of two integers m and n , it is easy to tell whether m divides n :

Theorem 11.14 Let $n = p_1^{a_1} \cdots p_k^{a_k}$ be a prime decomposition (i.e. p_i s are prime, $p_1 < \cdots < p_k$, and $a_i > 0$). Then m divides n if, and only if:

$$m = p_1^{b_1} \cdots p_k^{b_k}, \quad \text{with each } 0 \leq b_i \leq a_i. \quad (11.5.1)$$

Proof. We will prove each direction of this if and only if statement separately.

- ($\Leftarrow$): If (11.5.1) holds, then

$$n = m \cdot p_1^{a_1-b_1} \cdots p_k^{a_k-b_k} \implies m \text{ divides } n.$$

- ($\Rightarrow$): Suppose m divides n , then $n = mc$ for some positive integer c . Then m and c have prime decompositions whose product is the prime decomposition for n as shown below:

$$\underbrace{p_1^{a_1} \cdots p_k^{a_k}}_n = \underbrace{q_1^{c_1} \cdots q_l^{c_l}}_m \underbrace{r_1^{d_1} \cdots r_s^{d_s}}_c.$$

The FTA implies each q_i and r_i is equal to some p_j . To ease notation, we'll assume $q_i = r_i = p_i$ for all i , but that $c_i = 0$ if p_i didn't appear as one of the primes q_i originally, and similarly for the d_i . Then the product above is in fact

$$\underbrace{p_1^{a_1} \cdots p_k^{a_k}}_n = q_1^{c_1} \cdots q_l^{c_l} r_1^{d_1} \cdots r_s^{d_s} = p_1^{c_1+d_1} \cdots p_k^{c_k+d_k}.$$

The FTA now implies each power $c_i + d_i$ equals to a_i . In particular, this means $m = p_1^{c_1} \cdots p_k^{c_k}$ where $0 \leq c_k \leq a_k$. ■

Example 11.14 What are all the positive divisors of 360?

First, we find the prime factorization of 360: we can see that $36 = 4 \cdot 9 = 2^2 \cdot 3^2$, and so $360 = 10 \cdot 2^2 \cdot 3^2 = 2^3 \cdot 3^2 \cdot 5$. To list all the divisors, we just have to look at the values $2^j \cdot 3^k \cdot 5^\ell$ where $0 \leq j \leq 3$, $0 \leq k \leq 2$, and $0 \leq \ell \leq 1$. The possible powers of 2 are 1, 2, 4, 8, and the possible powers of 3 are 1, 3, 9, so we multiply the powers of two by these to get

$$\begin{array}{cccc} 1, & 2, & 4, & 8, \\ 3, & 6, & 12, & 24, \\ 9, & 18, & 36, & 72. \end{array}$$

The possible powers of 5 are just 1 and 5, so we can just multiply these numbers by 1 and 5 to get

$$\begin{array}{cccccccccccc} 1, & 2, & 4, & 8, & 3, & 6, & 12, & 24, & 9, & 18, & 36, & 72, \\ 5, & 10, & 20, & 40, & 15, & 30, & 60, & 120, & 45, & 90, & 180, & 360. \end{array}$$

11.6 LCM and GCD via prime factorizations

Definition 11.4 The least common multiple $\text{lcm}(a, b)$ of positive integers a and b is the smallest positive integer divisible by both a and b .

For example, $\text{lcm}(15, 12) = 60$.

Theorem 11.15 Let a and b have prime factorizations,

$$a = p_1^{r_1} \cdots p_m^{r_m}, \quad b = p_1^{s_1} \cdots p_m^{s_m}.$$

Here p_i 's are distinct, but r_i and s_i are allowed to be zero. Then:

- $\text{gcd}(a, b) = p_1^{\min(r_1, s_1)} \cdots p_m^{\min(r_m, s_m)}$.
- $\text{lcm}(a, b) = p_1^{\max(r_1, s_1)} \cdots p_m^{\max(r_m, s_m)}$.
- $\text{lcm}(a, b) = ab / \text{gcd}(a, b)$.

We leave the proof of this as an exercise.

Example 11.15 If $a = 120 = 2^3 \cdot 3 \cdot 5$ and $b = 36 = 2^2 \cdot 3^2$, then $\text{gcd} = 2^2 \cdot 3 = 12$ and $\text{lcm} = 2^3 \cdot 3^2 \cdot 5 = 360 = \frac{120 \cdot 36}{12}$.

Lemma 11.16 Let $r \in \mathbb{Q}$. Then there exist unique coprime integers a and b such that $r = \frac{a}{b}$ and $b \geq 1$.

Proof. Since r is rational, there are integers a' and b' such that $r = \frac{a'}{b'}$, with $b' \neq 0$. Without loss of generality, we may assume that $b' > 0$. If we find $-r = \frac{a}{b}$ with a and b coprime, then $r = \frac{-a}{b}$ with $-a$ and b coprime, so we may assume that $r \geq 0$.

Now let $p_1, \dots, p_k$ be the prime numbers that divide either a' or b' . We may therefore write

$$a' = p_1^{m_1} \cdots p_k^{m_k} \quad \text{and} \quad b' = p_1^{n_1} \cdots p_k^{n_k},$$

where each exponent m_i and n_i is nonnegative. Now for each i , let

$$s_i = \max(m_i - n_i, 0) \quad \text{and} \quad t_i = \max(n_i - m_i, 0).$$

Thus for each i , at most one of s_i and t_i is nonzero, and

$$\frac{p_i^{m_i}}{p_i^{n_i}} = \frac{p_i^{s_i}}{p_i^{t_i}}.$$

Thus we set

$$a = p_1^{s_1} \cdots p_k^{s_k} \quad \text{and} \quad b = p_1^{t_1} \cdots p_k^{t_k},$$

and we find that

$$r = \frac{a'}{b'} = \frac{a}{b}.$$

Now

$$\text{gcd}(a, b) = p_1^{\min(s_1, t_1)} \cdots p_k^{\min(s_k, t_k)} = p_1^0 \cdots p_k^0 = 1.$$

Now let us prove that a and b are unique with this property. If c and d are coprime integers with $d \geq 1$, and if $r = \frac{c}{d}$, then $ad = bc$. Since a and b are coprime, it follows that b divides d , and since c and d are coprime, it follows that d divides b . Thus $d = b$, and so $a = c$ as well. ■

11.7 Powers

Theorem 11.17 Let n be a positive integer. Then $\sqrt{n} \in \mathbb{Q}$ if and only if n is a perfect square, that is, $n = m^2$ for some integer m .

Proof. Suppose $\sqrt{n} \in \mathbb{Q}$. Thus $\sqrt{n} = \frac{a}{b}$, and $\frac{a^2}{b^2} = n$. By Lemma 11.16, we may assume that a and b are coprime and positive. Now since $n = \frac{a^2}{b^2}$, the uniqueness of Lemma 11.16 implies that $b^2 = 1$ and thus $n = a^2$ is a perfect square. Also, if $n = m^2$ for some integer m , then $\sqrt{n} = m = \frac{m}{1} \in \mathbb{Q}$. ■

Below, we say an integer a is an n th power if $a = b^n$ for some integer b .

Theorem 11.18 If $a, b \in \mathbb{N}$ are coprime, and ab is an n th power, then so are a and b .

Proof. If $a = 1$, then a is an n th power as $1 = 1^n$ and also we have $ab = b$ is an n th power, so the theorem is trivial in this case. Similarly, the theorem is true if $b = 1$. So we can assume $a, b \geq 2$, and apply the FTA.

Let $a = p_1^{r_1} \cdots p_k^{r_k}$ and $b = q_1^{s_1} \cdots q_\ell^{s_\ell}$ be the prime factorizations of a and b . Since a and b are coprime, they share no common prime factors, so $p_i \neq q_j$ for all j . By assumption $ab = c^n$ for some integer c . Let $w_1^{t_1} \cdots w_j^{t_j}$ be the prime factorization for c , then

$$ab = p_1^{r_1} \cdots p_k^{r_k} q_1^{s_1} \cdots q_\ell^{s_\ell} = c^n = w_1^{nt_1} \cdots w_j^{nt_j}.$$

By the FTA, these prime factorizations are equal, which means for each i , $p_i = w_j$ for some j , and $r_i = nt_j$. In particular, n divides r_i for all i , and so a is an n th power. The same holds for b . ■

Let's do an example of another diophantine equation.

Example 11.16 Find all integer solutions to $4x^2 = y^3 + 1$.

Reductions:

- If (x, y) is a solution, then $(-x, y)$ is also a solution, so we can assume $x \geq 0$.
- If $x = 0$, then $y^3 = -1$, which is only possible if $y = -1$. For any other solution, we can now assume $x > 0$. But then $y^3 + 1 = 4x^2 \geq 4$, so we must have $y > 0$ as well.

We are now left to finding all integer solutions $x, y > 0$. Observe that

$$y^3 = 4x^2 - 1 = (2x - 1)(2x + 1).$$

By The Linear Combination Lemma, the gcd of $2x - 1$ and $2x + 1$ must divide $2x + 1 - (2x - 1) = 2$, so the gcd is either 1 or 2. However, since both $2x - 1$ and $2x + 1$ are odd, the gcd must actually be 1. Hence, $2x \pm 1$ are coprime.

By Theorem 11.18, $2x \pm 1$ must both be cubes. Thus, there are $m, n \in \mathbb{Z}$ so that $m^3 = 2x + 1$ and $n^3 = 2x - 1$. Since x is a positive integer, so are m^3 and n^3 , and thus so are m and n . Then

$$2 = 2x + 1 - (2x - 1) = m^3 - n^3 = (m - n)(m^2 + mn + n^2),$$

so $m - n$ divides 2, hence it is either 1 or 2 (since $m > n$).

- If it is 1, then $m = n + 1$, and so by the above equation

$$2 = m^3 - n^3 = (n + 1)^3 - n^3 = 3n^2 + 3n + 1 \geq 3 \cdot 1^2 + 3 \cdot 1 + 1 = 7$$

which is a contradiction.

- If it is 2, then $m = n + 2$ and so

$$2 = m^3 - n^3 = (n + 2)^3 - n^3 = 6n^2 + 6n + 4 \geq 6 \cdot 1^2 + 6 \cdot 1 + 4 = 16$$

which is also a contradiction.

Thus, there are *no integer solutions* to $4x^2 = y^3 + 1$ when $x > 0$ and the only integer solution is $(x, y) = (0, -1)$.

11.8 Application: Pythagorean Triples

In this (non-examinable) section we will classify all Pythagorean Triples, that is, all positive integer solutions to

$$x^2 + y^2 = z^2.$$

This is a bit more involved than other diophantine problems and harder to figure out on your own (so it is not required reading), but we'll give a proof here since it's a neat application of what we've learned and can test your knowledge of the material in this chapter.

We narrow down the solutions in a few steps:

- If $z = 0$, then $x^2 = -y^2$ which is only possible if $x = y = 0$ and so we can assume $z > 0$.
- If $x = 0$, then we must have $y^2 = z^2$, and so $y = \pm z$. Thus, we know all solutions if $x = 0$, so let's assume $x > 0$. Similarly, we can assume $y > 0$.
- We can assume that x, y and z are coprime (that is, no two of them share a common factor other than 1). To see this, suppose $d = \gcd(x, y)$. Then $a = \frac{x}{d}$ and $b = \frac{y}{d}$ are coprime, and then

$$z^2 = x^2 + y^2 = d^2(a^2 + b^2).$$

By Theorem 11.18, $a^2 + b^2 = c^2$ for some integer c , and so $z^2 = d^2c^2 = (cd)^2$. Hence, $(x, y, z) = (da, db, dc)$ for some other Pythagorean Triple (a, b, c) where a and b are coprime. Thus, if we find all solutions (x, y, z) where x and y are coprime, then all other solutions are multiples of these. A similar proof shows that all solutions will be multiples of solutions where x and z are coprime and where y and z are coprime.

- In particular, assuming x, y , and z are coprime and positive solutions to $x^2 + y^2 = z^2$, then x or y is odd. We assume x is odd.
- We now claim $z + y$ and $z - y$ are coprime. Suppose not. Then there is a prime q that divides them both. But then

$$q \text{ divides } (z + y) - (z - y) = 2y, \quad q \text{ divides } (z + y) + (z - y) = 2z.$$

Since y and z are coprime, $q = 2$. But then

$$2 \text{ divides } (z - y)(z + y) = z^2 - y^2 = x^2$$

which implies x is even, a contradiction.

- Since $z \pm y$ are coprime and $x^2 = (z+y)(z-y)$, we know that $z+y = s^2$ and $z-y = t^2$ for some coprime integers s and t by Theorem 11.18 with $s > t$. Hence,

$$z = \frac{z+y+z-y}{2} = \frac{s^2+t^2}{2}$$

Similarly, $y = \frac{s^2-t^2}{2}$, and finally,

$$x^2 = (z-y)(z+y) = s^2t^2.$$

Thus, all positive coprime solutions are of the form

$$(x, y, z) = \left(st, \frac{s^2-t^2}{2}, \frac{s^2+t^2}{2} \right).$$

As we take x to be odd, then both s and t are also odd.

- Finally, we now recall that all solutions are multiples of these solutions, thus all Pythagorean Triples are of the form

$$(x, y, z) = \left(ast, a \frac{s^2-t^2}{2}, a \frac{s^2+t^2}{2} \right)$$

where a, s, t are integers.

We can now find the first 8 Pythagorean triples:

(x, y, z)	(t, s)	(x, y, z)	(t, s)	(x, y, z)	(t, s)	(x, y, z)	(t, s)
(3, 4, 5)	(1, 3)	(5, 12, 13)	(1, 5)	(7, 24, 25)	(1, 7)	(9, 40, 41)	(1, 9)
		(15, 8, 17)	(3, 5)	(21, 20, 29)	(3, 7)		
				(35, 12, 37)	(5, 7)	(45, 28, 53)	(5, 9)

11.9 Exercises

The relevant exercises in Liebeck's book are in Chapters 10 and 11.

- **Exercise 11.3** Show that n and $n^2 + n + 1$ are coprime.
- **Exercise 11.4** Find all $n \in \mathbb{N}$ so that $n-2$ divides n^2-2 .
- **Exercise 11.5** Show that $\gcd(n! + 1, (n+1)!)$ is either 1 or $n+1$ for all $n \in \mathbb{N}$.

Below, if an exercise just states an equation, find all integer solutions to that equation.

- **Exercise 11.6** $x^2 = 16y^2 + 8y + 2$
- **Exercise 11.7** $x^2 + 2y^2 = 8z + 5$.
- **Exercise 11.8** $x^3 - y^3 = 7$.
- **Exercise 11.9** $xy = x + y + 2y^2 - 1$.
- **Exercise 11.10** $x^3 - x = 12y + 6$.

- **Exercise 11.11** $xy + 2x + 3y = 4$.
- **Exercise 11.12** $6x^2 = 5x^3$.
- **Exercise 11.13 Challenging:** Find all integer solutions to the equation $x^3 + 3y^3 + 9z^3 = 0$.
- **Exercise 11.14** Find all primes p and integers $a, b \in \mathbb{N}$ so that $p^a - p^b = 24$.
- **Exercise 11.15** Is there a rational number q so that $q^5 - q^3 + 39 = 0$?
- **Exercise 11.16** Recall that a natural number n is *perfect* if it is the sum of all its proper divisors (that is, all positive divisors other than n). For example, $1 + 2 + 3 = 6$, so 6 is perfect. If p is a prime and $n \in \mathbb{N}$, when is p^n perfect?
- **Exercise 11.17** Show that $x^n + y^n = z^n$ has integer solutions if and only if it has rational solutions.
- **Exercise 11.18** Recall that there are infinitely many pythagorean triples, i.e. triples of positive integers (x, y, z) so that $x^2 + y^2 = z^2$. Using this, show that there are infinitely many integer solutions to $x^2 + y^2 = z^3$.
- **Exercise 11.19** Let

$$\mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} : a, b \in \mathbb{Z}\} = \{a + bi\sqrt{5} : a, b \in \mathbb{Z}\} \subseteq \mathbb{C}.$$

We say that $p \in \mathbb{Z}[\sqrt{-5}]$ is *primo* if the only solutions to $p = ab$ with $a, b \in \mathbb{Z}[\sqrt{-5}]$ have either a or b equal to ± 1 .

- (a) Prove that $|a + b\sqrt{-5}| = \sqrt{a^2 + 5b^2}$. Conclude that if $|z| = 1$, then $z = \pm 1$.
 - (b) Assuming (a), prove that every nonzero $z \in \mathbb{Z}[\sqrt{-5}]$ can be written as a product of primos.
 - (c) Show that $2, 3, 1 \pm \sqrt{-5}$ are all primo.
- **Exercise 11.20** Show that if $n > 4$ is an integer that is *not* prime, then n divides $(n-1)!$.
 - **Exercise 11.21** Recall that the Fermat numbers $F_n = 2^{2^n} + 1$ satisfy the recurrence relation

$$F_n = F_{n-1} \cdots F_0 + 2.$$

Show that the Fermat numbers are mutually relatively prime, that is, $\gcd(F_n, F_m) = 1$ whenever $m \neq n$.

11.9.1 Challenging Exercise: A special case of Fermat's last theorem*

In this exercise we will show that $x^4 + y^4 = z^4$ (the Fermat quartic) has no *nontrivial solutions*, that is, no integer solutions apart from $(x, y, z) = 0$. The method we will use is known as “infinite descent”: we suppose there is a solution (x, y, z) with $z > 0$, then we assume (x, y, z) has z smallest among all such solutions, then we show that we can find another solution (u, v, w) with $w < z$, contradicting that z was the smallest integer appearing in a solution

to $x^4 + y^4 = z^4$. This is quite difficult and wouldn't be expected of you on a homework or exam, it's just a bit of fun.

- Explain why it suffices to only find positive solutions?
- Why does it suffice to show $x^4 + y^4 = z^2$ has no nontrivial integer solutions?
- Suppose there is a non-trivial solution positive solution to $x^4 + y^4 = z^2$. Then there is a solution (x, y, z) with smallest $z > 0$ among all solutions. Show that x, y, z have no common factors.
- Show that exactly one of x or y is odd. Now assume x is odd and y is even.
- Note that $a = x^2$ and $b = y^2$ along with $c = z$ form a Pythagorean triple, a is odd, and x, y, z are coprime. Hence, by our work on classifying Pythagorean triples, we must have

$$(a, b, c) = \left(as^2t^2, a \frac{s^2 - t^2}{2}, a \frac{s^2 + t^2}{2} \right)$$

for some $s > t$. Show that 4 divides $s - t$.

- Using that $(s - t)(s + t) = 2b = 2y^2$, show that there are integers u and v with $s + t = 2u^2$ and $s - t = 4v^2$, and u and $2v$ are relatively prime. solving for s and t , verify that $x^2 + 4v^4 = u^4$.
- Let $A = x$, $B = 2v^2$ and $C = u^2$. Show that this is another Pythagorean triple of the form described in the parametrization theorem. So they can be expressed as above for (a, b, c) , but with parameters S and T instead of s and t .
- Arguing in a similar fashion to (d), show that S and T satisfy $S + T = 2X^2$ and $S - T = 2Y^2$ for some positive integers X and Y .
- Solve for S and T and verify that $u^2 = X^4 + Y^4$. Finally show that $uXY \neq 0$ and that $0 < u < z$, leading to a contradiction.